

Find the Percent of a Number

Lesson 4-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*30% means 30 out of 100***Percent**

A way to compare a number to 100, shown with the % sign.

*In '25% of 80,' the base is 80***Base**

The whole amount you take a percent of.

*25% of 80 = 20; the part is 20***Part**

The piece you get when you take a percent of a number.

*part = percent $\times$ base***Equation**

A math sentence with an equal sign showing both sides are the same.

*\$40 shirt at 25% off: discount = \$10***Discount**

Money taken off the first price to make it cheaper.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find the percent of a number using an equation or a model.

1 Notice

It's Discount Day at the arcade!

2 Key idea

I can find the percent of a number using an equation or a model.

3 Words I'll use

Percent

Base

Part

Equation

Discount



Think About It

- What information do you need to find the discount amount?
- How is finding 25% of 60 different from finding 40% of 80?
- Is there more than one way to find a percent of a number?



Watch

Watch your teacher model one example. Jot what you see.



We try

Solve the next one together as a class.



You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

On Discount Day the claw machine is 25% off its 60-ticket price. In '25% of 60,' which number is the base (the whole), and what part are you trying to find?

Level 1 support

Start here: The base is 60 tickets and the part is the 25% discount we need to find.

 **Need a hint?**

Sentence starters (optional)

The base is ____ and the part I want is ____.

La base es ____ y la parte que quiero es ____.

I need to find ____% of ____.

Necesito encontrar el ____% de ____.

Word bank: percent base part discount

Listen for: Students identify 60 as the base, the 25% discount as the part to find, and connect 'of' to the base.

Level 2

Without computing exactly, will the discount be more or less than half of 60? Use a benchmark to estimate and explain.

- The discount will be ____ than half because ____.
- I estimate about ____ tickets because ____.

EXPLORE

What steps did you follow every time to find the percent of a number in the discount table?

Level 1 support

Start here: I changed the percent to a decimal, then multiplied it by the base to get the part.

 **Need a hint?**

Sentence starters (optional)

I changed ____% into the decimal ____.

Cambié ____% al decimal ____.

Then I multiplied ____ x ____ = ____ (the part).

Luego multipliqué ____ x ____ = ____ (la parte).

Word bank: percent base part equation

Listen for: Students describe the equation $\text{part} = \text{percent} \times \text{base}$, converting the percent to a decimal first (e.g., $0.40 \times 80 = 32$).

Level 2

Compare finding 25% of 60 with the decimal 0.25 versus dividing 60 into 4 equal parts.

Why do both methods give the same part?

- Both give ____ because 25% is the same as ____.
- I prefer the ____ method because ____.

CONNECT

The arcade donates 12% of Saturday's 1,500 ticket sales. The manager says 180 tickets, a student says 18. Who is right, and how do you know?

Level 1 support

Start here: 12% of 1,500 is 0.12 times 1,500, which is 180 tickets.

 Need a hint?

Sentence starters (optional)

The ____ is correct because 12% of 1,500 = ____ x 1,500 = ____.

El/La ____ tiene razón porque 12% de 1,500 = ____ x 1,500 = ____.

The mistake was using ____ instead of ____.

El error fue usar ____ en lugar de ____.

Word bank: percent base part equation

Listen for: Students compute $0.12 \times 1,500 = 180$ and explain the student likely used 0.012 (divided by 1,000) instead of 0.12.

Level 2

Estimate first: 10% of 1,500 is easy to find. Use that to check whether 180 is reasonable for 12%, and explain your reasoning.

- 10% of 1,500 is ____, so 12% should be a little ____.
- 180 is reasonable because ____.

Guided Examples

Example 1

What is 20% of 150?

Solution: $20\% = 0.20$. Multiply: $0.20 \times 150 = 30$.

Answer: A. 30

Example 2

What is 50% of 84?

Solution: $50\% = 0.50$. Multiply: $0.50 \times 84 = 42$. (Or simply divide by 2: $84 \div 2 = 42$.)

Answer: A. 42

Example 3

What is 10% of 320?

Solution: $10\% = 0.10$. Multiply: $0.10 \times 320 = 32$. (Or move the decimal one place left.)

Answer: A. 32

Write About the Math

The Writing Revolution

I can explain my steps using the words percent, base, part, and equation.

1. Kernel Sentence subject + verb

Model: Percent is a way to compare a number to 100, shown with the % sign.

Porcentaje es una manera de comparar un número con 100, con el signo %.

Write a kernel sentence about percent. Use a subject and a verb.

Escribe una oración base sobre porcentaje. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Percent matters in math

Porcentaje importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Percent matters in math because ____.

Porcentaje importa en matemáticas porque ____.

but
pero

Percent matters in math, but ____.

Porcentaje importa en matemáticas, pero ____.

so
entonces

Percent matters in math, so ____.

Porcentaje importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about percent.

Di un hecho verdadero sobre percent.

Percent ____.

Question *Pregunta*

Ask a question about percent.

Haz una pregunta sobre percent.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about percent.

Muestra entusiasmo sobre percent.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with percent.

Dile a un compañero qué hacer con percent.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The base is ____ and the part I want is ____.

La base es ____ y la parte que quiero es ____.

I need to find ____% of ____.

Necesito encontrar el ____% de ____.

I changed ____% into the decimal ____.

Cambié ____% al decimal ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. An arcade prize costs 240 tickets. If you have a coupon for 15% off, how many tickets do you save?

- A. 36 tickets
- B. 24 tickets
- C. 15 tickets
- D. 204 tickets

Show your work:

2. A student scored 72 out of 90 on a test. What percent did they score?

- A. 80%
- B. 72%
- C. 75%
- D. 90%

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A class has 32 students. 25% are in the math club, 50% are in the reading club, and 12.5% are in both. How many students are in each group? Can the total of all three groups be more than 32? Explain why or why not.

Sentence starter: There are ____ in math club, ____ in reading club, and ____ in both. The total ____ be more than 32 because ____.

Show your work:

Reflect — Exit Ticket

What is 45% of 360?

- A. 162
- B. 145
- C. 180
- D. 126

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 36 tickets — $15\% \text{ of } 240 = 0.15 \times 240 = 36 \text{ tickets saved.}$
2. **Try It 2:** A. $80\% - 72 \div 90 = 0.80 = 80\%.$
3. **Exit Ticket:** A. $162 - 45\% = 0.45.$ *Multiply:* $0.45 \times 360 = 162.$

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Percent is a way to compare a number to 100, shown with the % sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).